

What is Programming

In []: Programming **is** the process of giving instructions to a computer to perform specific task.

What is Python?

In []: Python **is** high level , interpreted , **object-oriented** **and** general purpose programming language

History of Python

In []: Python was created by Gaido van Rossum.
Developement **1989**
Python **0.9.0** was released **1991**

In [4]: **a=10**
b=20
c=a+b
print(c)
print(b)

30
20

In [5]: **a=10**
b=20
c=a+b
c

Out[5]: 30

Why Learn Python?

In []: Easy to learn
Easy ro write
Easy to understand
Powerful
Versatile
Widely used **in** Industry

Why Companies Prefer Python

In []: Faster Developement
Less code
Huge collection of Libraries
Large Community Support
Easy maintance

Applications of Python

Web Development

Framework

```
In [ ]: Django
        Flask
        FastAPI
```

```
In [ ]: Example:
        Instagram
        Spotify
```

Machine Learning

Application

```
In [ ]: Recommendation System
        Fraud Detection
        Spam Detection
        Price Prediction
```

Libraries

```
In [ ]: Scikit-learn
        Tensoeflow
        PyTorch
```

Artificial Intelligence

```
In [ ]: Chatbots
        Voice Assistants
        Image Recognition
        Natural Language Processing
```

```
In [ ]: Examples:
        Gemini
        ChatGPT
        Claude
```

Cyber Security

```
In [ ]: Ethical Hacking
        Network Automation
        Security Testing
```

Desktop Applications

```
In [ ]: GUI applications can be built using:  
        Tkinter  
        PyQt  
        Kivy
```

Game Developement

```
In [ ]: Pygame
```